

Inferential Provenance for Geometry–Dynamics Alignment

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Abstract

Geometry–dynamics alignment claims often compress a ranking or selected statistic into an explanation without preserving the chain of custody that made the claim interpretable. A leading Lyapunov/Oseledets direction may be dynamically important without being the direction in which a declared local stability landscape is most curved, and a selected maximum over modes is not the same claim as a predeclared fixed-mode comparison. This paper proposes an audit framework for such claims. The framework treats the alignment spectrum—the projection-mass representation of a declared Hessian direction or unstable Hessian subspace over a declared Lyapunov/Oseledets frame—as an enabling object for provenance-preserving compression, not as evidence by itself. The reporting chain is Object $\rightarrow$ Representation $\rightarrow$ Functional $\rightarrow$ Selection Rule $\rightarrow$ Reference/Robustness $\rightarrow$ Claim. The worked U.S. Treasury-market example is deliberately subordinate to that chain: it shows how conclusions change when full-frame mass, selected modes, references, and robustness are reported together. The example illustrates an audit contract; it does not establish a fixed mode, a policy prescription, or a mechanism.

1 Introduction: Inferential Provenance for Alignment Claims

Many high-dimensional diagnostics begin by asking for the largest object. In a covariance analysis this is the first principal component [Jolliffe, 2002]. In a factor model it is the dominant factor [Bai and Ng, 2002]. In a local dynamical analysis it is the leading Lyapunov or Oseledets direction [Oseledets, 1968, Benettin et al., 1980, Arnold, 1998]. In each case the ranking has a clear meaning: it orders variance, factor strength, or finite-time perturbation growth. The deeper error is not simply choosing the wrong leading mode. It is compressing a ranking or statistic into an explanation without a visible chain of custody from object definition to interpreted claim.

This paper is about that chain of custody. Let the local object be a scalar stability, loss, or stress landscape and let its second-order geometry be summarized by a Hessian direction or unstable Hessian subspace. Let the dynamical object be a Lyapunov/Oseledets frame extracted from the realized evolution of the state. A direction can be dynamically important without being the direction in which the local landscape is most curved. A direction can be geometrically important without being the leading dynamical mode. More importantly, a scalar summary of their relationship can lose the decisions that make the summary interpretable: what object was measured, how it was represented, which functional was applied, whether the coordinate was fixed or selected, what reference geometry was used, how robust the result was, and what claim was finally allowed.

The contribution is an audit framework for geometry–dynamics alignment claims: a provenance-preserving compression framework built around the geometry–dynamics coupling measure. It answers a methodological question:

When a local geometric object and a Lyapunov/Oseledets frame are both available, what provenance must be preserved before a compressed alignment statistic can be interpreted?

The answer is the reporting chain

Object $\rightarrow$ Representation $\rightarrow$ Functional $\rightarrow$ Selection Rule $\rightarrow$ Reference/Robustness $\rightarrow$ Claim.

The spectrum is sufficient for this reporting problem because it keeps the full projection-mass distribution over the declared dynamical frame before scalar summaries are taken. It is not asserted to exhaust all geometry–dynamics phenomena.

The individual ingredients are intentionally standard: projection masses, Parseval’s identity, principal-angle/subspace comparisons, Haar reference geometry, multiple-testing logic, and robustness maps. What this paper adds is their enforced order. Alignment is first represented as a measure over the declared dynamical frame; only then are fixed-mode, selected-mode, weighted, entropy, reference-divergence, and subspace functionals reported. The selection rule determines the reference comparison, robustness is read as sensitivity information rather than proof, and the final claim grade controls how much domain language the statistic is allowed to carry.

The boundary is narrow. This is not a new estimator, Lyapunov-vector algorithm, forecasting model, or guarantee of empirical truth. Auditability is not validity. The protocol is conditional on the declared representation and metric, so it is not invariant to arbitrary reparameterization. Robustness strengthens interpretation only by exposing which claims survive declared perturbations and which do not. Its uncertainty layers are object, representation, selection, and interpretation; finite-sample Hessian and Lyapunov-frame uncertainty remain future extensions. Mode labels may also move across specifications while a mode family or subspace remains the more stable object of interpretation.

The bond-market application supplies a stress test for the framework. Treasury and monetary-policy data are a natural setting because yield-curve factors, policy surprises, volatility, and regime changes are already studied with low-dimensional summaries and dynamic tools [Jolliffe, 2002, Bai and Ng, 2002, Nelson and Siegel, 1987, Diebold and Li, 2006, Kuttner, 2001, Gürkaynak et al., 2005, Acosta et al., 2025, Corsi, 2009, Hamilton, 1989]. Those literatures make it tempting to ask which single mode matters. The audit framework asks a different question: after object, representation, functional, selection rule, reference, and robustness choices are made visible, what claim is still licensed by the full coupling measure over the declared dynamical frame? The empirical section is therefore a demonstration of operational consequences, not a claim of universal necessity or a Treasury-market mechanism.

The paper makes five contributions about provenance-preserving reporting:

1. **Object and representation.** It defines the alignment spectrum as the coordinate representation of a coupling measure from a declared Hessian direction or unstable Hessian subspace to a declared Lyapunov/Oseledets frame or dynamical subspace.
2. **Functional discipline.** It separates the full coupling measure from fixed-mode, selected-mode, weighted, entropy, reference-divergence, and subspace summaries, so no scalar is mistaken for the object itself.
3. **Selection provenance.** It distinguishes predeclared fixed coordinates from posthoc selected maxima, ensuring that selected labels are not reported as if they had been fixed in advance.
4. **Reference and robustness.** It matches the null/reference geometry to the selection rule and maps the coupling measure across declared specifications, including metric, bandwidth, target, state definition, and estimator choices.
5. **Claim licensing.** It assigns ROBUST, CONDITIONAL, PATTERN, or UNVALIDATED grades before empirical patterns are promoted into domain language.

The rest of the paper follows the provenance chain. Section 2 gives context. Section 3 separates the local geometric object from the realized dynamical frame. Section 4 defines the alignment spectrum and its functionals. Section 5 turns the decomposition into an auditable reporting interface. The empirical sections use the Treasury-market state panel only as a worked example of how the interface changes what can be said.

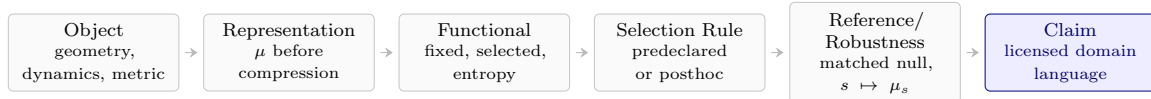


Figure 1: Inferential-provenance chain for geometry–dynamics alignment. Each reported claim inherits its license from the declared object, representation, functional, selection rule, and matched reference/robustness record.

2 Related Work and Context

This paper sits between three bodies of work, but it is not a substitute for any of them. The first is the theory and computation of finite-time dynamical modes. Oseledets’ multiplicative ergodic theorem provides the asymptotic frame for Lyapunov exponents and invariant directions [Oseledets, 1968]; QR and Benettin-style product methods make exponent computation operational in finite samples [Benettin et al., 1980]; and Lyapunov-vector recovery algorithms clarify the distinction between orthonormal computational vectors and dynamically covariant directions [Ginelli et al., 2007, Wolfe and Samelson, 2007]. Random dynamical-systems treatments clarify that these objects are properties of evolution operators, not of a static loss surface [Arnold, 1998]. The present paper uses that distinction as a reporting constraint: a Lyapunov ranking is a dynamical ranking, not a Hessian ranking. A reviewer may also view the protocol as an interface between finite-time Lyapunov-vector computation and subspace comparison. The reporting contribution here is not a new estimator, but the requirement that the estimator output be treated as a full frame or declared subspace before any mode label is interpreted. Subspace statements should be tied to principal angles rather than to arbitrary basis labels [Björck and Golub, 1973, Stewart and Sun, 1990].

The second context is empirical bond-market modeling. Yield-curve work often compresses the Treasury curve into level, slope, and curvature factors [Nelson and Siegel, 1987, Diebold and Li, 2006]. Monetary-policy event studies and high-frequency surprise measures provide disciplined ways to connect policy news to asset-price movements [Kuttner, 2001, Gürkaynak et al., 2005, Acosta et al., 2025]. Volatility and connectedness tools add dynamic summaries of financial stress [Corsi, 2009, Diebold and Yilmaz, 2012]. Those tools are useful inputs and comparators. They do not by themselves answer whether the geometric direction identified by a local Hessian is the leading dynamical direction, a subdominant direction, a subspace, or no stable direction at all.

The third context is causal and policy caution. Empirical policy relationships can change when the rule or environment changes [Lucas, 1976], and modern quasi-experimental designs emphasize explicit identifying assumptions, robustness checks, and sensitivity to specification choices [Callaway and Sant’Anna, 2021, Goodman-Bacon, 2021, Rambachan and Roth, 2023, Montiel Olea and Plagborg-Møller, 2021]. This paper borrows that discipline at the level of measurement: alignment claims require declared objects, declared selection rules, reference geometry, and falsification conditions. The protocol is therefore closer to a reporting standard than to a new structural model.

A final context is loss-landscape and Hessian geometry. Hessian eigenvectors, eigengaps, and negative-curvature subspaces are routinely used to summarize local geometry in optimization and learning systems [Nocedal and Wright, 2006, Sagun et al., 2017, Ghorbani et al., 2019]. The present paper imports only the reporting problem: if local curvature and realized dynamics are both measured, their comparison needs an object declaration, a metric declaration, and a selection rule.

3 Geometry and Dynamics Are Different Objects

Let $x \in \mathbb{R}^d$ denote a standardized system state. A local stability, loss, or stress landscape is a scalar field

$$V : \mathbb{R}^d \rightarrow \mathbb{R}, \quad x \mapsto V(x).$$

Its second-order local geometry is the Hessian

$$H(x) = \nabla^2 V(x).$$

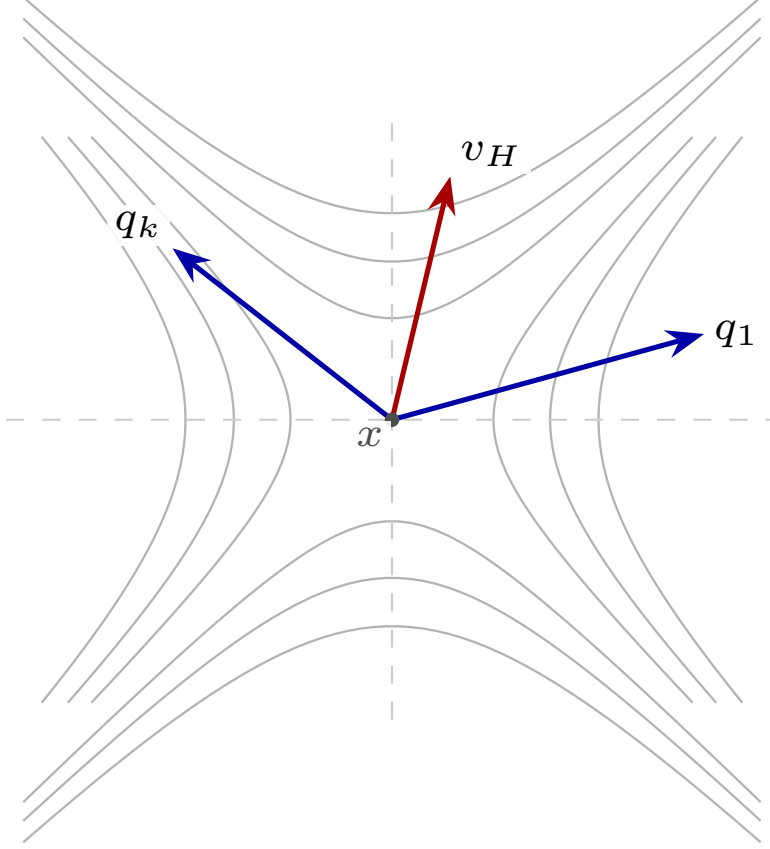


Figure 2: Local curvature is a geometric object, while Lyapunov directions are dynamical objects. The schematic shows a local saddle landscape: the Hessian direction v_H is a curvature direction, while q_1 and q_k are declared dynamical directions. The alignment problem is not to assume q_1 is geometrically relevant, but to report how curvature mass projects across the declared dynamical frame.

Given a positive-definite metric $M \succ 0$, a unit direction is measured by $\|v\|_M^2 = v^\top M v$. A vector-level Hessian object can be defined as the direction of most negative curvature,

$$v_H(x) \in \arg \min_{\|v\|_M=1} v^\top H(x) v.$$

This is the Rayleigh-quotient form of a negative-curvature direction; second-order optimization conditions make the negative Hessian eigenspace the relevant local-instability object [Nocedal and Wright, 2006]. If the negative part of the Hessian is separated and one-dimensional, this single-vector description is meaningful. If several negative eigenvalues are near each other, or if the eigengap is not established, the geometric object is better reported as an unstable subspace U_H rather than a labeled vector.

The dynamical object comes from an evolution law, not from the Hessian alone. In discrete time write

$$X_{t+1} = F_t(X_t) + \varepsilon_{t+1}, \quad J_t = DF_t(X_t).$$

For a horizon h , the tangent map is

$$\Phi_{t,h} = J_{t+h-1} \cdots J_{t+1} J_t.$$

Finite-time Lyapunov exponents are obtained from the singular values of this product,

$$\lambda_i(t, h) = \frac{1}{h} \log \sigma_i(\Phi_{t,h}), \quad \sigma_1 \geq \cdots \geq \sigma_d,$$

or from an equivalent QR product computation [Oseledets, 1968, Benettin et al., 1980, Arnold, 1998]. The corresponding forward modes are written $q_1(t), \dots, q_d(t)$ and are ordered by dynamical importance.

Geometry side. A scalar landscape $V(x)$ gives a Hessian $H(x)$ and a declared geometric object: a direction v_H or unstable subspace U_H .

Dynamics side. The realized evolution gives Jacobian products $\Phi_{t,h}$ and a declared dynamical object: a frame $Q = [q_1, \dots, q_d]$ or subspace Q_r .

Reporting interface. Only after the metric $M \succ 0$ is declared does the paper report the coupling measure $\mu = \sum_i p_i \delta_i$ or its subspace analogue. Equality between the two sides is not assumed.

Figure 3: Generic object map for the alignment protocol. The figure intentionally avoids a dense arrow diagram: local geometry and realized dynamics are separate inputs, and the coupling report is a metric-dependent comparison between declared objects.

The distinction is elementary but consequential. The Hessian answers a local curvature question about V . The Lyapunov frame answers a realized-evolution question about products of Jacobians. These objects can coincide in special systems, for example local gradient-flow benchmarks where the Jacobian is tied to $-H$. Outside such benchmarks, equality is an assumption. The purpose of the alignment protocol is to make that assumption visible and testable.

3.1 Vector and subspace versions

The vector version asks how a declared Hessian direction v_H projects onto a declared dynamical frame $Q = [q_1, \dots, q_d]$. The subspace version asks how an unstable Hessian subspace U_H overlaps a declared dynamical subspace Q_r . Both are legitimate alignment objects, but they should not be mixed in prose. The representation step is therefore deliberately separated from the object choice: object, representation, functional, selection rule, reference geometry, and final claim are distinct layers of the report. A vector-level statement says that a particular direction has a large coordinate in a particular mode. A subspace-level statement says that two spans have small principal angles or large projection overlap [Björck and Golub, 1973]. When exact mode labels are unstable, the subspace statement is often the safer object.

4 The Coupling Measure over the Dynamical Frame

The coupling measure is not a single angle. It is a measure of how geometric mass is distributed over a declared dynamical frame. Let $g = v_H / \|v_H\|_M$ be a metric-unit Hessian direction. Let $E_1, \dots, E_d$ denote the declared one-dimensional dynamical subspaces associated with the reported Lyapunov/Oseledets frame, and let Π_i^M be the M -orthogonal projector onto E_i . When the frame is represented by M -orthonormal vectors $q_1, \dots, q_d$,

$$\Pi_i^M z = q_i \langle q_i, z \rangle_M, \quad \|q_i\|_M = 1, \quad \langle q_i, q_j \rangle_M = 0 \quad (i \neq j).$$

If the reported Lyapunov vectors are not an M -orthogonal resolution of the state space, the mass interpretation below should not be used without first stating the orthogonalization or subspace projection convention. In that case, the report is a full alignment-score vector of raw normalized pairwise cosines, not a probability-valued coupling measure; its squared entries need not sum to one. This caveat is important for covariant or finite-time Lyapunov representatives: their dynamical meaning does not by itself make them an orthogonal resolution.

Definition 1: geometry–dynamics coupling measure. Given a metric-unit geometric direction g and a declared M -orthogonal dynamical resolution $E_1 \oplus \dots \oplus E_d = \mathbb{R}^d$, define

$$p_i(g; Q, M) = \|\Pi_i^M g\|_M^2 = |\langle g, q_i \rangle_M|^2, \quad i = 1, \dots, d.$$

The geometry–dynamics coupling measure is the probability measure on mode labels

$$\mu_{g,Q,M} = \sum_{i=1}^d p_i(g; Q, M) \delta_i.$$

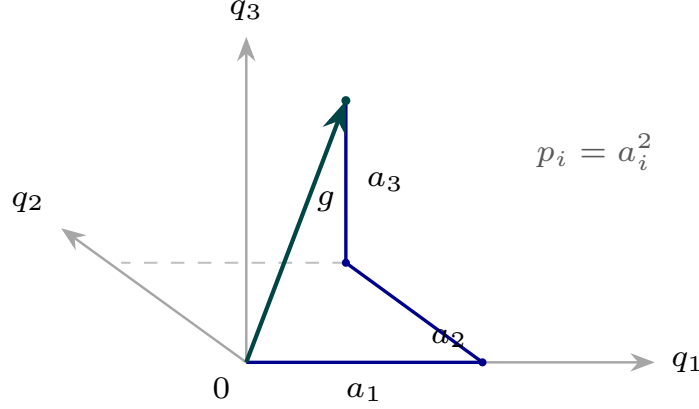


Figure 4: Schematic of the coupling measure as projection mass over a declared dynamical frame. The vector g represents the metric-unit Hessian direction, the axes represent declared M -orthonormal dynamical directions, and the squared projection lengths give the masses $p_i = a_i^2$.

Equivalently, $p = (p_1, \dots, p_d)$ is a probability vector over declared mode labels. The alignment spectrum is the square-root coordinate representation

$$a_i(g; Q, M) = \sqrt{p_i(g; Q, M)} = \frac{|\langle v_H, q_i \rangle_M|}{\|v_H\|_M \|q_i\|_M}, \quad \mathcal{A} = (a_1, \dots, a_d).$$

Thus the spectrum is not merely a list of cosines; after a metric and an orthogonal frame convention are declared, it is a distribution of geometric mass over dynamical labels. For the reporting problem in this paper, that distribution is a sufficient representation for the fixed-mode, selected-mode, weighted, entropy, effective-support, divergence, and declared frame-subspace functionals below. It is not a universal representation of every temporal, sign, pathwise, or phase-sensitive geometry-dynamics relation.

Proposition 1: basic structural property. If $q_1, \dots, q_d$ form a complete M -orthonormal frame and g is metric-unit, then

$$0 \leq p_i \leq 1, \quad \sum_{i=1}^d p_i = 1.$$

Moreover, p_i is invariant to sign changes of g or q_i and to transformations that preserve the declared metric. It is not invariant to an arbitrary rescaling of state coordinates.

Proof. The projectors Π_i^M are mutually orthogonal and resolve the identity, so Parseval's identity in the inner product $\langle u, v \rangle_M = u^\top M v$ gives $\|g\|_M^2 = \sum_i \|\Pi_i^M g\|_M^2$. Since $\|g\|_M = 1$, the masses sum to one. Nonnegativity and the upper bound are immediate. Sign invariance follows from squaring the coordinate. Metric dependence follows because both the unit sphere and the projectors are defined by M .

Dispersion summaries. Scalar summaries are functionals of the coupling measure, not replacements for it. A concentration-sensitive summary is the Shannon entropy

$$H(\mu_{g,Q,M}) = - \sum_{i:p_i>0} p_i \log p_i,$$

with effective support

$$N_{\text{eff}}(\mu_{g,Q,M}) = \exp\{H(\mu_{g,Q,M})\}.$$

Here $1 \leq N_{\text{eff}} \leq d$. Values near one mean that the geometric object is concentrated on a small number of dynamical labels; values near d mean that the mass is broadly spread. This is a descriptive geometric summary only. It is not a mechanism, a forecasting statement, or a policy statement.

When a reference mass distribution $\rho = (\rho_1, \dots, \rho_d)$ is declared with $\rho_i > 0$ and $\sum_i \rho_i = 1$, a descriptive divergence can also be reported:

$$D_{\text{KL}}(\mu_{g,Q,M} \parallel \rho) = \sum_{i:p_i>0} p_i \log \frac{p_i}{\rho_i}.$$

The safest default reference is the uniform distribution $\rho_i = 1/d$. If a Haar or Monte Carlo reference is used, the manuscript must state whether the comparison is to fixed-coordinate tails, selected-max tails, or the expected mass vector. The KL number is a scale-dependent descriptive divergence from a declared reference, not a p-value and not evidence for a market-generating null.

Fixed and selected functionals. A fixed-mode claim and a selected-mode claim are different functionals of the same measure. For a predeclared mode k ,

$$T_k(\mu) = p_k.$$

For a selected maximum,

$$T_{\max}(\mu) = \max_{1 \leq i \leq d} p_i, \quad m^* \in \arg \max_{1 \leq i \leq d} p_i.$$

For declared weights $w_i \geq 0$ with $\sum_i w_i = 1$, a weighted report is

$$T_w(\mu) = \sum_{i=1}^d w_i p_i.$$

These functionals answer different questions. T_k asks whether mass falls on a predeclared dynamical label. $T_{\max}$ asks how much mass can be found after searching across labels. T_w asks how much mass falls in a declared weighted region of the frame. Reporting $T_{\max}$ as if it were T_k hides selection. Reporting T_w without the weights hides the geometry being summarized.

Proposition 2: scalar insufficiency is an information-loss property. Except in degenerate designs where all unreported coordinates are irrelevant by assumption, no one-coordinate scalar $T_k(\mu) = p_k$ and no selected-maximum scalar $T_{\max}(\mu) = \max_i p_i$ preserves the fixed-mode, selected-mode, entropy/effective-support, weighted, and declared subspace-relevant information in μ simultaneously. Two coupling measures can share the same principal coordinate, or share the same selected maximum, while differing on the remaining mass allocation and hence on other admissible reports.

Proof. For a one-coordinate report, fix any value $c \in [0, 1]$ for p_k . The remaining mass $1 - c$ can be placed on one coordinate, spread uniformly over the other coordinates, or arranged according to declared weights. These choices leave T_k unchanged while changing $T_{\max}$, H , N_{eff} , T_w , and the mass assigned to any reported family or coordinate subspace that intersects the unreported coordinates. For a selected-maximum report, when the frame has at least three relevant labels, one can keep the maximum value and its selected label fixed while redistributing mass among nonmaximal labels. That redistribution leaves $T_{\max}$ unchanged but changes fixed-mode reports, entropy/effective support, weighted summaries, and family or subspace masses. Therefore neither scalar identifies the coupling measure or the collection of protocol functionals derived from it.

Subspace and Grassmann extensions. When the Hessian object is an unstable subspace rather than a stable single vector, let $U \subset \mathbb{R}^d$ be a p -dimensional Hessian-unstable subspace and let Π_U^M denote its M -orthogonal projector. The coupling measure over the dynamical frame is

$$p_i(U; Q, M) = \frac{1}{p} \text{tr}(\Pi_i^M \Pi_U^M), \quad \mu_{U,Q,M} = \sum_{i=1}^d p_i(U; Q, M) \delta_i.$$

If $E_1 \oplus \dots \oplus E_d = \mathbb{R}^d$ is a complete M -orthogonal resolution, then $\sum_i p_i(U; Q, M) = 1$. This reports how an unstable Hessian subspace spreads across individual dynamical labels, without pretending that a single basis vector inside U is canonical.

For comparison with a declared r -dimensional dynamical subspace $R \subset \mathbb{R}^d$, use the basis-free projection overlap

$$A_{\text{sub}}(U, R; M) = \frac{1}{\min(p, r)} \text{tr}(\Pi_U^M \Pi_R^M) = \frac{1}{\min(p, r)} \sum_{j=1}^{\min(p, r)} \cos^2 \theta_j,$$

where θ_j are the principal angles between U and R in the declared metric [Björck and Golub, 1973, Edelman et al., 1998, Absil et al., 2008]. A corresponding projection distance is

$$d_{\text{proj}}^2(U, R; M) = \|\Pi_U^M - \Pi_R^M\|_{F, M}^2 = p + r - 2 \text{tr}(\Pi_U^M \Pi_R^M),$$

with the Frobenius norm taken after mapping to M -orthonormal coordinates. Subspace statements should use these Grassmann-type objects rather than labeled vectors whenever eigengaps or Lyapunov-mode separations do not support exact mode names. If labels migrate across adjacent modes but a declared family or span remains stable, the report should name the family or subspace rather than treating label migration itself as semantic instability.

The reporting rule is therefore: first report the coupling measure $\mu_{g, Q, M}$ or $\mu_{U, Q, M}$; then, if useful, report fixed, selected, weighted, entropy, divergence, or subspace functionals with their selection rule and reference geometry declared. The Treasury application below is a worked example of this reporting discipline, not evidence for a fixed mode or a mechanism.

5 Inferential Provenance Framework for Reporting Alignment

The reporting framework treats an alignment claim as an inferential object with provenance. The checklist is only its implementation form. Its purpose is not to create a new estimator or to make the empirical alignment more valid than the data allow. Its purpose is to make the chain from geometry to interpretation auditable: hidden defaults become declarations, selected summaries are not mistaken for fixed tests, and robustness choices cannot silently lend false strength to a fragile result.

The framework is needed because geometry–dynamics alignment contains several places where a paper can change the object without changing the displayed number. The geometric object may be a vector or a subspace; the dynamical representation may be a full finite-time frame or a declared mode family; the reported functional may be a full spectrum, a fixed coordinate, a selected maximum, or a subspace overlap; and the reference comparison must match the selection rule. Transparency and auditability are therefore distinct from validity. A fully audited report can still be wrong, sample-specific, or substantively unimportant. The framework does not strengthen the claim; it prevents undisclosed representation, selection, and robustness choices from making the claim look stronger than it is.

The minimum complete chain is

$$\text{Object} \rightarrow \text{Representation} \rightarrow \text{Functional} \rightarrow \text{Selection Rule} \rightarrow \text{Reference/Robustness} \rightarrow \text{Claim}.$$

Each layer is necessary. The object says what local geometric entity is being compared. The representation states the metric, standardization, and dynamical frame in which an angle has meaning. The functional declares whether the report uses the full coupling measure, a coordinate, a selected maximum, a weighted summary, a divergence, or a subspace comparison. The selection rule determines which reference distribution is relevant. The reference and robustness layers separate orientation-scale calibration from specification sensitivity. The claim layer then limits the interpretation to what survived the declared chain.

5.1 Declare the object

State whether the geometric object is a Hessian direction, a set of negative-curvature Hessian directions, an unstable Hessian subspace, or another declared local object. State how the object is estimated and whether its label is supported by an eigengap or stability check. If the object is a subspace, avoid naming a single vector as if it were canonical. State whether the dynamical object is the full Lyapunov/Oseledets frame, a finite-time frame at a declared horizon, a mode family, or a declared dynamical subspace. Mode labels can move across specifications even when mode families or subspaces remain the meaningful object.

5.2 Declare the representation

Angles are representation-dependent statements. The paper must report the state-space metric, standardization, or whitening convention under which $\langle \cdot, \cdot \rangle_M$ and $\| \cdot \|_M$ are computed, together with the ordering convention and estimator used to construct the dynamical frame. If alternative standardizations or horizons are credible, they belong in the robustness map. A change in metric can change the alignment spectrum, so the metric is part of the result rather than a nuisance convention.

5.3 Declare the functional

Report the coupling measure

$$\mu_{g,Q,M} = \sum_{i=1}^d p_i \delta_i, \quad p_i = a_i^2,$$

or its subspace analogue before reporting any scalar summary. If a scalar summary is useful, define it as a functional of the measure: a fixed coordinate T_k , a selected maximum $T_{\max}$, a weighted average T_w , an entropy/effective-support summary, a declared-reference divergence, or a Grassmann subspace overlap. Such summaries help compare specifications, but they do not replace the distribution of geometric mass over the declared dynamical frame. For this reporting problem, the spectrum is sufficient to make mode migration and concentration auditable; it is not claimed to exhaust all geometry–dynamics phenomena.

Case studies may report raw alignment-score vectors $\mathcal{A}$ for interpretability. Label them diagnostic: conservation, null-reference calibration, and claim-grading logic attach to the mass representation $p_i = a_i^2$ under a declared orthogonal projection convention, not to unnormalized raw scores.

5.4 Declare the selection rule

A fixed-mode statement should use language such as “for the predeclared mode q_k .” A selected-mode statement should use language such as “the selected maximum over declared modes.” The manuscript should not move between these forms without saying so. The leading mode may be included, but it is not the only candidate unless that restriction is itself the stated research design. The null or reference comparison must match the selection rule.

5.5 Attach reference geometry and robustness

A reference geometry asks how large an alignment could be under orientation alone. For an isotropic unit vector u and a fixed unit vector q in dimension d ,

$$|\langle u, q \rangle|^2 \sim \text{Beta} \left(\frac{1}{2}, \frac{d-1}{2} \right).$$

This follows from the standard spherical/Dirichlet representation of squared coordinates of a Haar-uniform direction [Muirhead, 1982]. That is a fixed-mode reference. A selected maximum over a frame requires the max-over-coordinates reference induced by the joint projection masses, not only the fixed-mode tail. This reference is a diagnostic benchmark; it is not a claim that the empirical system is generated by an isotropic or GOE market model. The distinction is the geometric analogue of reporting fixed tests separately from post-selection maxima [Westfall and Young, 1993, Romano and Wolf, 2005].

Robustness is the companion provenance layer. Let $s \in \mathcal{S}$ index bandwidth, target definition, state variables, sample window, standardization, and dynamics estimator. Each specification produces

$$\mathcal{A}_s = (a_{s1}, \dots, a_{sd}), \quad m(s) \in \arg \max_i a_{si}.$$

The robustness object is the map

$$s \mapsto \mathcal{A}_s,$$

not a single preferred realization. A stable conclusion should survive across the declared grid or be explicitly limited to the part of the grid where it holds. If $m(s)$ moves across specifications, mode identity should be described as specification-dependent, and the paper should foreground the spectrum, mode family, or subspace rather than the selected label.

5.6 Grade the claim before interpreting it

The final report should assign grades before interpretation. A ROBUST claim is one that survives the declared specification map and directly supports the paper-facing conclusion. A CONDITIONAL claim is an in-sample or design-limited witness and must carry its conditions in the sentence. A PATTERN claim reports a repeated empirical shape while leaving mechanism open. An UNVALIDATED claim is quarantined transparency material and should not be written as a result.

5.7 Implementation checklist

Table 1: Implementation checklist for the inferential provenance framework

Framework layer	Required report	Failure prevented
Object	Hessian direction, unstable Hessian subspace, Lyapunov/Oseledets frame, or declared dynamical subspace	Hiding whether the claim is vector, subspace, or mode-family level
Representation	State-space metric, standardization, frame estimator, and mode ordering	Incomparable angle claims
Functional	Full mass distribution μ , raw score vector $\mathcal{A}$, fixed coordinate, selected maximum, or subspace overlap	One-coordinate summaries
Selection rule	Predeclared fixed mode or posthoc selected maximum	Fixed/selected-mode conflation
Reference/robustness	Fixed-mode and selected-mode reference tails plus specification map	Unpriced selection and one-realization claims
Claim	ROBUST / CONDITIONAL / PATTERN / UNVALIDATED	Mechanism upgrades

6 Robustness and Selection Rules

Robustness is not an appendix to the claim. It is part of the claim object. A single score vector or coupling measure states what happened under one declared specification; the robustness map states whether the reported relationship survives the credible choices around metric, bandwidth, target, sample, state variables, and dynamics estimator. This matters because the selected coordinate $m(s)$ can move even when the broader lesson remains stable.

This convention deliberately prioritizes transparency over discovery efficiency. Requiring declared grids and selection-aware references can reduce the apparent power of a study, and it may understate fragile but substantively important signals that appear only under a narrow specification. That is a Type-I/Type-II tradeoff in reporting form: the framework is designed to avoid overstating selected evidence, not to maximize the chance of finding every possible alignment pattern.

A fixed-mode robustness claim asks whether the predeclared coordinate remains interpretable across the grid. A selected-mode robustness claim asks whether the winner and its magnitude are stable after selection is accounted for. A full-frame robustness claim asks whether the reported score vector or mass measure remains informative even as the selected label changes. These claims should not be collapsed into one sentence. This fixed-versus-selected distinction follows the same logic as multiple-testing and data-snooping adjustments: the reference distribution must match the search rule [Westfall and Young, 1993, White, 2000, Romano and Wolf, 2005].

The empirical application below uses this distinction directly. The robust paper-facing claims are F1 and F2: principal-mode summaries are insufficient in this application, and alignment should be reported across the full spectrum and with bandwidth disclosed. The identity of a selected coordinate is not treated as a stable finding unless the robustness map supports that stronger statement.

In the measure-valued formulation, robustness is a map

$$s \mapsto \mu_s, \quad \mu_s = \sum_{i=1}^d p_{si} \delta_i,$$

when an orthogonal projection convention supports a mass interpretation. When the empirical object is a raw alignment-score vector, the same reporting rule applies to $s \mapsto \mathcal{A}_s$ instead. The selected label $m(s)$ can move even when the broader score pattern remains qualitatively similar; conversely, the same selected label can hide a large change in dispersion or subspace overlap. For that reason, mode migration should be reported together with the underlying score vector, mass distribution, or subspace object.

7 Null Geometry as a Reference, Not a Market Model

Null/reference geometry belongs in the protocol because it separates an angle from the surprise attached to that angle. A selected maximum has searched across a frame. A fixed coordinate has not. A reporting protocol that treats those two objects as the same will overstate selected-mode evidence.

The caveat comes before the numbers. The Haar reference is a reference orientation, not a market model and not a validation device. It does not say that the Treasury state is isotropic, that the observed spectrum is generated by GOE structure, or that a small tail would establish a mechanism. It only prices the difference between reading a predeclared coordinate and reading the largest coordinate after searching the frame.

Let u be Haar-uniform on the unit sphere S^{d-1} and let q be any fixed unit vector. Then

$$Z = |\langle u, q \rangle|^2 \sim \text{Beta} \left(\frac{1}{2}, \frac{d-1}{2} \right).$$

Thus a fixed-mode tail is

$$\mathbb{P}(Z \geq z) = 1 - I_z \left(\frac{1}{2}, \frac{d-1}{2} \right),$$

where I_z is the regularized incomplete beta function. The beta law is the standard one-coordinate marginal of a Haar-uniform spherical direction [Muirhead, 1982]. This is the correct law for a predeclared fixed mode such as q_1 .

Selected-mode claims require a different null. In an orthonormal basis,

$$(|\langle u, q_1 \rangle|^2, \dots, |\langle u, q_d \rangle|^2) \sim \text{Dirichlet} \left(\frac{1}{2}, \dots, \frac{1}{2} \right).$$

[Muirhead, 1982]. The selected statistic

$$M_d = \max_i |\langle u, q_i \rangle|^2$$

must therefore be compared to the max-over-modes null, or at least to a multiplicity-adjusted fixed-mode tail. In the current implementation, the selected tail is estimated by Monte Carlo and Bonferroni fixed-mode tails are recorded as a conservative reference.

The orientation reference below is a separate specification from the bandwidth-1.0 spectrum in Table 3. It is a null-diagnostic reference, not the conditional q_4 witness used in Section 9. For $d = 6$, this separate specification has fixed-mode alignment 0.163 with analytic fixed-mode tail 0.727 and Monte Carlo tail 0.715; its selected-mode alignment is 0.663 with max-over-modes Monte Carlo tail 0.623. These numbers are not evidence of a market-isotropy theorem. They show that selected-mode reads must pay a selection cost.

This reference also ignores estimation dependence between $\hat{H}$ and $\hat{Q}$. The reported selected-tail values are finite-sample Monte Carlo reference quantities whose precision depends on the simulation design and seed; they do not account for finite-sample uncertainty in $\hat{H}$, in the finite-time Lyapunov frame, or in the alignment functional built from them. The null should therefore be read as an orientation-scale diagnostic for fixed-versus-selected reporting, not as a p-value for a data-generating null. Bootstrap procedures or uncertainty propagation through Hessian estimation and Lyapunov-frame construction are natural future extensions, but they are not part of the Haar reference used here.

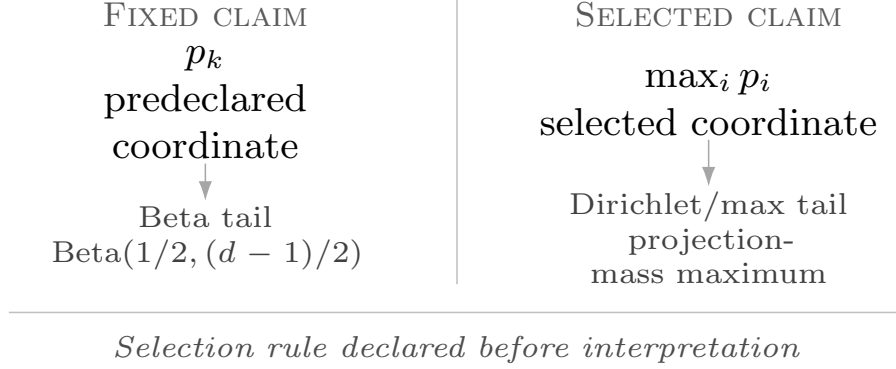


Figure 5: Fixed-mode and selected-mode alignment claims use different reference objects. A predeclared coordinate is compared with the fixed-mode tail; a posthoc maximum is compared with the max-over-modes tail. The diagram illustrates the selection rule and reference comparison, not a market-generating model.

The GOE connection is limited. GOE eigenvectors are Haar-oriented under rotational invariance [Mehta, 2004], so the Haar null is a useful reference orientation. But this is not a GOE spectrum or outlier test, and it does not require the market data to be generated by a GOE model.

Table 2: Fixed-mode and selected-mode null/reference geometry

Selection rule	Observed alignment	Squared alignment	Reference law	Analytic tail	MC tail	Adjustment
Predeclared fixed q_1	0.163	0.027	Beta(1/2, (d - 1)/2)	0.727	0.715	none
Posthoc max over modes	0.663	0.440	Dirichlet(1/2, ..., 1/2)	–	0.623	max-over-modes
Selected threshold as fixed mode	0.663	0.440	Beta(1/2, (d - 1)/2)	0.104	–	fixed-threshold reference
Bonferroni fixed-mode reference	0.663	0.440	fixed-mode tail $\times d$	–	0.627	conservative multiplicity reference

The reference geometry uses the standardized Euclidean state-space metric, the mode labels q_1 through q_6 , random seed 20260603, and 1,000 random-geometry iterations. The selected maximum has a much smaller fixed-mode tail at the same threshold than its max-over-modes tail; this is exactly why the selection rule must be stated before interpreting the number.

The same caution applies to divergence summaries. A number such as $D_{\text{KL}}(\mu||\rho)$ compares an observed coupling measure with a declared reference mass distribution ρ . It does not replace the fixed-mode beta tail or the selected max-over-modes reference, because those tails answer selection-aware questions about functionals of the random projection vector. Thus a divergence summary may describe concentration relative to a reference, but it should not be written as a market-model p-value or as evidence for a mechanism.

8 Empirical Design: U.S. Treasury-Market State as Protocol Testbed

The empirical testbed is a standardized six-dimensional U.S. Treasury-market state panel with 1,600 daily rows from 2020-01-02 through 2026-06-01, built from Treasury yields, effective federal funds rates, macro stress inputs, SPY equity data, and the full yield curve. The application is used to demonstrate the operational consequences of ignoring provenance in geometry–dynamics alignment reports. It is not offered as a proof of universal necessity, a Treasury-market mechanism, or a structural state model. The reported empirical alignment values are raw alignment scores: absolute cosines between the Hessian curvature object and dynamics modes in the declared standardized Euclidean metric. They are diagnostic summaries, not treated as probability masses unless an orthogonal frame convention is separately declared; single-vector Hessian reports also need the eigengap/stability condition. The inputs mirror standard yield-curve, monetary-surprise, volatility, and regime-state summaries [Nelson and Siegel, 1987, Diebold and Li, 2006, Kuttner, 2001, Gürkaynak et al., 2005, Acosta et al., 2025, Corsi, 2009, Hamilton, 1989].

The empirical design separates a baseline spectrum from a robustness map. The robustness grid has 504 predeclared specifications and varies bandwidth, center, dynamics estimator, sample window, standardization, target, and variable set. Of those specifications, 468 complete successfully and 36 fail or are skipped. The paper-facing use of this grid is not to select a single winning index or prove that Treasury dynamics must have a particular channel; it is to show what interpretive claims change when object, representation, functional, selection rule, and robustness provenance are kept visible.

For reproducibility, each empirical report must specify four estimator choices: the scalar target used to fit $\hat{V}$, the local smoothing/bandwidth rule used before taking $\nabla^2 \hat{V}$, the dynamics estimator used to form the finite-time tangent map, and the rule by which specifications are marked failed or skipped. In this application those details are treated as part of the specification index s , not as nuisance implementation choices. Tables below therefore report only claims that can be traced to source artifacts and to the declared grid; auditability makes the claim inspectable, not automatically valid.

The protocol is applied in six provenance-preserving steps:

1. Declare the object: estimate a local stability landscape $\hat{V}(X)$ and its Hessian $H = \nabla^2 \hat{V}$ under a declared bandwidth.
2. Declare the representation: use the stated state vector, standardization, and Euclidean metric rather than treating coordinates as canonical.
3. Declare the functional: extract the Hessian curvature direction or subspace, estimate the Lyapunov/Oseledets frame $q_1, \dots, q_6$ under the declared dynamics estimator, and compute $\mathcal{A} = (a_1, \dots, a_6)$.
4. Declare the selection rule: distinguish fixed-mode from selected-mode claims before interpreting any coordinate or maximum.
5. Declare the reference and robustness evidence: attach the appropriate reference geometry and repeat over the robustness grid.
6. Declare the claim: grade the evidential status before adding domain interpretation.

9 Empirical Demonstration: Baseline Alignment Spectrum

The baseline bandwidth-1.0 score vector is the simplest demonstration of the operational cost of reporting only a dominant or preselected coordinate. The q_1 coordinate is 0.021, while the largest coordinate is 0.621 on q_4 . These are raw alignment scores, not a normalized mass distribution. The latter is a conditional in-sample witness: it shows that an off-principal coordinate can matter in this declared specification, but it does not by itself license a stable mode-identity claim or a Treasury mechanism.

Table 3: Baseline Hessian–Lyapunov alignment-score vector at bandwidth 1.0

Mode	Lyapunov exponent	$ \cos(v_H, q_i) $
q_1	−0.00177	0.021
q_2	−0.00177	0.052
q_3	−0.00812	0.083
q_4	−0.01460	0.621
q_5	−0.03434	0.357
q_6	−0.05717	0.432

Weighted summaries compress the score vector only after the full vector has been reported and remains recoverable. At bandwidth 1.0, the selected-mode value is 0.621, while weighted summaries range from 0.261 under uniform weights to 0.416 under energy weights. Weighting choices emphasize different parts of the same alignment vector, so the scalar is a convenience report and not a replacement for (0.021, 0.052, 0.083, 0.621, 0.357, 0.432) or for the fixed q_1 /selected maximum distinction.

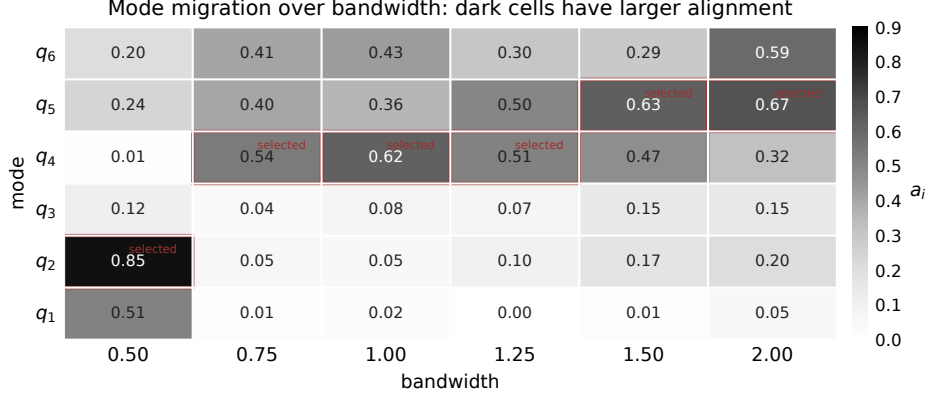


Figure 6: Mode-migration heatmap for the bandwidth-sensitivity table. Cell tone reports the raw alignment score a_i ; thin red outlines mark the selected coordinate at each bandwidth. The display replaces the earlier 3D surface so the table values remain directly comparable without perspective distortion.

10 Empirical Demonstration: Robustness and Mode Migration

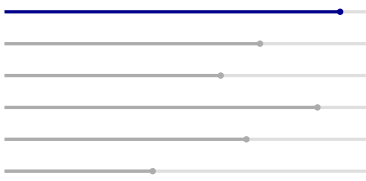
Bandwidth moves both the estimated Hessian eigenvalue and the selected alignment coordinate. Across the six bandwidths $\{0.50, 0.75, 1.00, 1.25, 1.50, 2.00\}$, the selected mode is q_2 at bandwidth 0.50, q_4 at bandwidths 0.75, 1.00, and 1.25, and q_5 at bandwidths 1.50 and 2.00. That $q_2/q_4/q_5$ migration is label-instability evidence, not evidence that the economy switches among three named mechanisms. Mode-4 alignment has mean 0.411, standard deviation 0.221, minimum 0.007, and maximum 0.621 over this bandwidth sweep. If economically named mode families or subspaces remain stable while individual labels move, interpretation should move from point labels to semantic or subspace stability. The observed subdominant, credit-spread-loaded mode is therefore a conditional in-sample witness, not a stable channel claim.

Table 4: Bandwidth sensitivity and selected-mode migration

Bandwidth	a_1	a_2	a_3	a_4	a_5	a_6	Selected mode	Hessian eigenvalue
0.50	0.512	0.849	0.118	0.007	0.240	0.205	q_2	-161.9
0.75	0.015	0.054	0.035	0.537	0.404	0.405	q_4	-55.3
1.00	0.021	0.052	0.083	0.621	0.357	0.432	q_4	-73.6
1.25	0.001	0.101	0.071	0.511	0.501	0.301	q_4	-30.8
1.50	0.015	0.172	0.151	0.474	0.626	0.286	q_5	-21.1
2.00	0.046	0.205	0.149	0.318	0.669	0.593	q_5	-18.1

The specification multiverse supports the protocol claim directly. Among the 468 successful specifications, q_1 is the strongest coordinate in 104 cases, or 22.2% of the completed grid. A non-principal coordinate is strongest in 364 cases, or 77.8%. The selected coordinate is distributed across all six modes rather than concentrated in a single index. The consequence is evidential, not mechanistic: selection-aware provenance changes the status of the claim one is allowed to make.

Table 5: Strongest-mode distribution across the robustness multiverse

Mode	Share of successful specifications	Count	Share
q_1		104	22.2%
q_2		79	16.9%
q_3		67	14.3%
q_4		97	20.7%
q_5		75	16.0%
q_6		46	9.8%

Across successful cells, the selected maximum has mean 0.458 and median 0.444, while the fixed q_1 coordinate has mean 0.242 and median 0.183. These distributional summaries reinforce the protocol point: selected maxima and fixed q_1 alignments answer different questions, so compressing them into one headline number would erase the provenance needed for interpretation.

11 What Survives in This Application

The empirical application leaves two claims safe to put in the foreground. First, principal-mode summaries are insufficient in this application. Second, alignment should be reported across the full spectrum with enough provenance to recover the object, representation, functional, selection rule, reference, and robustness surface behind the claim. These claims do not require a stable winning mode. They rest on the failure of the single-mode reduction and on the movement of mode identity across the reported specification grid.

The strongest coordinate at bandwidth 1.0 is useful as a witness rather than as a destination. It shows why provenance-preserving compression matters, but it does not license a stable mode story. A subdominant, credit-spread-loaded mode is a conditional in-sample witness, not a stable channel claim. The claim grade stays CONDITIONAL because the interval evidence is wide, the mode index moves across bandwidths, and the same family has not been locked across all robustness axes. If a family or subspace were stable while its label changed, the paper-facing interpretation would be semantic/subspace stability rather than label identity.

The predictive layer is weaker still. Alignment features have medium-horizon predictive content in some specifications; timing is distributed and metric-dependent, and mechanism remains open. That sentence is the complete paper-facing F4 claim. It is not a forecasting-model claim, and it is not a mechanism claim.

Table 6: Claim grades and falsification rules

Claim	Grade	Paper-facing language	What would falsify or upgrade it
F1	ROBUST	Principal-mode summaries are insufficient in this application.	Falsified if q_1 remains strongest and sufficient across bandwidths, samples, components, targets, estimators, and reconstruction nulls.
F2	ROBUST	Alignment should be reported across the full spectrum and with bandwidth disclosed.	Falsified if full-spectrum reporting adds no information beyond q_1 and bandwidth disclosure does not change interpretation.
F3	CONDITIONAL	Conditional in-sample witness.	Upgraded only if bootstrap intervals exclude near-zero across designs and the same credit-loaded family dominates across samples and components.
F4	PATTERN	Pattern; mechanism open.	Falsified if alignment features fail to improve out-of-sample metrics across medium horizons, stress definitions, and non-degenerate regimes.
F5	UNVALIDATED	Insufficient evidence; quarantine-only.	Upgraded only if a predeclared alignment series, spectral estimator, and frequency band show sign-stable calm/crisis low-frequency differences across estimator and sample checks.

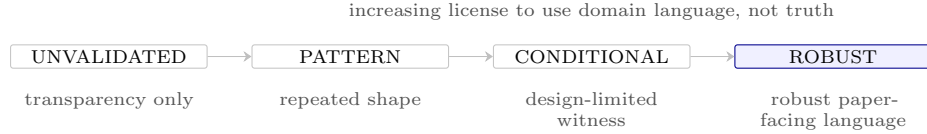


Figure 7: Claim-status licensing hierarchy used by Table 6. The scale governs paper-facing language, not evidential truth.

Table 6 is part of the protocol rather than an after-the-fact disclaimer. It prevents three common upgrades: converting a selected in-sample coordinate into a stable mode claim, converting a predictive pattern into a mechanism, and converting robustness evidence into scientific importance. Robustness changes evidential status by exposing sensitivity; it is not a guarantee of truth.

The predictive layer is retained only as an example of claim grading. It asks whether alignment features add medium-horizon information in rolling-origin checks after baseline information is already included, using predictive-comparison logic rather than a production forecasting claim [Diebold and Mariano, 1995, Giacomini and White, 2006]. Alignment features have medium-horizon predictive content in some specifications; because the specification search is visible, interpretation remains subject to data-snooping and superior-predictive-ability cautions [White, 2000, Hansen, 2005]. The distributed-lag check at horizon 21 with lags 0–10 gives $R^2 = 0.135$, cumulative effect -0.142 , and concentration ratio 0.234, so timing is reported as distributed and metric-dependent. This is the full paper-facing F4 claim: PATTERN, mechanism open.

12 Limitations, Failure Modes, and Falsification

The protocol is designed to make limitations visible. Its first limitation is representation dependence. Alignment is an angle only after the state-space metric or standardization is declared. A different standardization changes the inner product and can change the reported spectrum. Nonlinear reparameterizations can change the representation more deeply; the protocol is conditional on the declared representation and is not reparameterization-invariant. This is not a nuisance; it is part of what the paper asks authors to report.

The second limitation is mode-label instability. Single eigenvectors are meaningful only when the relevant eigengaps are meaningful. If Hessian eigenvalues or Lyapunov exponents are close, individual labels can rotate under small perturbations even when a subspace is stable. In that case, the safer object is semantic or subspace alignment, as in Section 4.

The third limitation is selection and performativity. A predeclared fixed-mode claim and a selected maximum are different statistical objects. Fixed-mode claims can use the fixed-mode reference geometry. Selected maxima must pay the search cost through a max-over-modes reference or a multiplicity-adjusted fixed-mode tail. A high selected coordinate is not automatically surprising, and a public robustness checklist can be gamed if authors search until a claim looks stable. The evidential object is the declared grid and its failures, not the existence of many checks.

The fourth limitation is empirical scope. The U.S. Treasury-market state is a worked state for the reporting protocol, not a canonical state vector for monetary systems. The sample window, stress target, state components, standardization, and dynamics estimator all belong in the reported specification surface.

The fifth limitation is scale. A paper need not print every intermediate array in the main text, but compression must remain provenance-preserving: the reader must be able to recover the object, representation, functional, selection rule, reference, robustness surface, and claim grade behind the reported scalar or table. Otherwise compression becomes another way to hide selection.

The sixth limitation is uncertainty layering. The application exposes object, representation, selection, and interpretation uncertainty, but it does not provide a full finite-sample uncertainty theory for the Hessian estimate or the Lyapunov frame. A fully calibrated version of the framework would propagate uncertainty through the entire chain

$$\hat{V} \rightarrow \hat{H} \rightarrow \hat{v}_H \rightarrow \hat{Q} \rightarrow \hat{A},$$

including eigengap-sensitive uncertainty for Hessian and Lyapunov subspaces, bootstrap or subsampling variability of the alignment vector, and selection-adjusted intervals for fixed and selected functionals. The present paper does not solve that calibration problem. It supplies the provenance structure such calibration would need: the object, representation, functional, selection rule, reference, robustness surface, and claim must be declared before uncertainty statements can be interpreted. This is why the empirical section remains a demonstration of operational consequences rather than a validity proof.

The falsification rule is deliberately stronger than a single counterexample. F1 would fail if q_1 remained strongest and sufficient across the declared robustness grid. F2 would fail if the full spectrum and bandwidth disclosure added no information. F3 would upgrade only under stable interval and family evidence across designs. F4 would fail if the alignment features stopped adding out-of-sample information across the medium-horizon, stress-definition, and non-degenerate-regime checks. F5 remains outside the result set.

The low-frequency spectral mechanism remains exploratory and unvalidated because current estimator sweeps reverse sign and the tested object is dominant/ q_1 rolling alignment, not q_4 -specific evidence. This limitation does not weaken the protocol claim: F1 and F2 do not depend on spectral evidence.

13 Generalization Beyond Finance: Reporting Rules for Other Systems

The protocol transfers when a study has two objects: a local geometric object and a dynamical object. The domain changes; the reporting discipline does not. The examples below are transfer targets, not validated case studies; they apply the same mathematical separation between local geometry, tangent or rollout dynamics, metric choice, and subspace comparison developed above [Nocedal and Wright, 2006, Oseledets, 1968, Björck and Golub, 1973, Edelman et al., 1998].

In control, cost or Lyapunov-function curvature can point in a different direction from closed-loop perturbation modes. In robotics, value or policy-loss geometry can identify locally fragile directions that do not match the largest rollout mode. In representation learning, Hessian, Fisher, or gradient-feature geometry can differ from training drift or recurrent-state modes. In scientific machine learning, reduced-model residual geometry can differ from operator or simulator modes. In neuroscience and physics, high-variance evolution directions can differ from task- or instability-relevant geometry.

The common template is: declare the object, declare the representation, declare the functional, declare the selection rule, attach reference and robustness evidence, and state the claim before adding domain language.

The same template also answers follow-up questions without opening new result sets. If the largest coordinate is not q_1 , the protocol asks whether the coordinate was fixed in advance or selected after looking across the frame. If the selected coordinate moves, it reports $s \mapsto \mu_s$ rather than averaging away the movement. If labels move while a family or subspace remains coherent, interpretation follows the stable semantic/subspace object rather than the unstable index. If predictive or spectral patterns appear, it keeps them separate from mechanism unless the claim grade supports an upgrade.

14 Conclusion: Audit the Provenance, Preserve the Compression

The technical question is not whether the largest mode is important. The technical question is how to compare geometric importance with dynamical importance without letting a convenient scalar hide the object, representation, functional, selection rule, reference geometry, robustness surface, or claim grade that produced it.

The answer is provenance-preserving compression. The alignment spectrum

$$\mathcal{A} = (a_1, \dots, a_d),$$

is the basic report because it keeps the frame visible, but the durable contribution is broader than “show the spectrum.” A scalar summary can be useful only when its provenance remains recoverable: the metric, selection rule, robustness map, reference geometry, and claim grade must travel with it. In the Treasury-market application, the principal-mode summary is insufficient and the strongest coordinate changes across specifications. That demonstrates the operational consequence of hiding provenance; it does not prove a universal law or identify a Treasury mechanism.

The conclusion is intentionally narrow. The paper does not identify a fixed mode, propose a production forecasting model, issue a policy prescription, or validate a spectral mechanism. It gives a reusable way to say what was aligned, under which representation, by which functional and selection rule, against which reference, across which specifications, and at which claim grade. That is auditability, not validity; it makes interpretive upgrades harder to smuggle into compressed empirical reports.

A Data State, Standardization, and Metric Declaration

The empirical state is a six-dimensional standardized Treasury-market and monetary-policy panel. The components summarize policy-rate conditions, inflation/credibility information, credit-spread information, equity/market conditions, growth information, and yield-curve shape. The current multiverse run covers 1,600 daily observations from 2020-01-02 through 2026-06-01.

All alignment values are computed in the declared standardized state-space metric M . A change in standardization changes the meaning of angle and can change the alignment spectrum. The state vector is therefore part of the reported empirical surface. It is not asserted to be canonical or exhaustive.

Table 7: Empirical grid and data inputs

Item	Reported value
Loaders	<code>load_treasury_yields</code> , <code>load_effr</code> , <code>load_macro_stress_panel</code> , <code>load_equity('SPY')</code> , <code>load_full_curve</code>
State panel builder	<code>mgov.policyeval.build_state_panel</code>
Sample window	2020-01-02 through 2026-06-01; $n = 1,600$ rows
Variable sets	base6; growth plus policy; no policy rate; no credit proxy; nominal only
Dynamics methods	var1; var2; local-linear confirmation slice
Targets	stress realized volatility; stress composite
Standardization	true/false grid
Random seed	20260603
Default dynamics stride and minimum count	stride 5; minimum count 80

B Robustness Multiverse and Mode-Migration Inventory

The mode-migration counts are reported in Table 5. The appendix records the reading rule: the robust object is the map from specification to spectrum, with bandwidth, state components, target definition, sample window, standardization, and dynamics estimator kept visible. The key fact is not that one non-principal mode wins; it is that the winning index is distributed across the frame. The $q_2/q_4/q_5$ bandwidth migration in the main text is therefore read as label instability. If economic mode families or subspaces are stable, the interpretation should be attached to those families or subspaces rather than to a migrating label.

C Bandwidth, Bootstrap, and Conditional Witness Checks

Table 8: Bandwidth and interval checks for the conditional witness

Check	Value	Claim-safety reading
Bandwidth grid	six values from 0.5 to 2.0	The reported coordinate depends on smoothing.
Mode-4 alignment over grid	mean 0.411, std. 0.221, min 0.007, max 0.621	The index is not bandwidth-invariant.
Bootstrap point and mean	point 0.621, mean 0.458	In-sample witness only.
Bootstrap dispersion	std. 0.197, 95% CI [0.066, 0.758]	Wide uncertainty; near-zero values remain in the interval.
Near-orthogonal null check	$p = 0.054$	Keep the grade CONDITIONAL.

A subdominant, credit-spread-loaded mode is a conditional in-sample witness, not a stable channel claim. This appendix explains why the main text uses the witness only to motivate full-spectrum reporting.

D Null Geometry: Fixed-Mode and Selected-Mode Reference Tails

The appendix uses the same separate null-diagnostic specification reported in Table 2. The key point is the fixed-versus-selected distinction, not a second empirical finding: a fixed coordinate uses the one-coordinate beta tail, while a selected maximum uses the max-over-frame reference. Haar orientation is a reference orientation, not a GOE market model; the null geometry is not a GOE spectrum test, not an outlier test, and not a market isotropy theorem.

E Predictive Exploration Details

Alignment features have medium-horizon predictive content in some specifications; timing is distributed and metric-dependent, and mechanism remains open. The predictive appendix records supporting diagnostics without turning the manuscript into a forecasting paper. The comparison is interpreted with standard predictive-accuracy and conditional-predictive-ability cautions [Diebold and Mariano, 1995, Giacomini and White, 2006].

Table 9: Predictive exploration checks retained as bounded diagnostics

Check	Reported value	Claim-safety reading
Distributed-lag check	horizon 21, lags 0–10, $R^2 = 0.135$	Predictive content is summarized as a pattern.
Lag accumulation	cumulative effect -0.142 , concentration ratio 0.234	Timing is distributed.
Metric-stability horizon maxima	best horizon differs by scoring rule	Timing is metric-dependent.
Rolling-origin and stress-definition checks	cited as F4 artifacts	Mechanism remains open.

The F4 artifacts are retained because they test whether spectrum features add information in some out-of-sample configurations. They do not define a production forecasting model.

F Spectral Quarantine and Unvalidated Mechanisms

This appendix is transparency material. It is not part of the paper’s result set. The low-frequency spectral mechanism remains exploratory and unvalidated because current estimator sweeps reverse sign and the tested object is dominant/q1 rolling alignment, not q4-specific evidence.

Table 10: Spectral quarantine summary

Quarantine item	Recorded value	Interpretation
What was tested	dominant/q1 rolling alignment	Not q4-specific spectral evidence.
Welch baseline p-value	0.006	Detectable under one baseline, but not enough to validate F5.
Positive calm-minus-crisis estimators	Welch(128), Multitaper	Some estimators point one way.
Sign-reversal estimators	Welch(64), Welch(256), Periodogram	Estimator choice changes the sign.
Decision rule	appendix-only when sign reverses or p-value is not significant	Keep F5 UNVALIDATED / quarantine.

The quarantined spectral material is included so that a tested but insufficient mechanism does not disappear. It should not be cited as support for F1 or F2, and it should not be promoted into the abstract, introduction, or conclusion.

G Reproducibility Note

The empirical case is backed by a research stack rather than a production trading system. Four artifact layers matter for this paper. Data inputs are kept under `data/`; analysis code and state-space routines live under `src/py/mgov/`; follow-up analyses and verification scripts live under `scripts/`; and paper-facing CSV, JSON, markdown, and figure outputs live under `results/`. The manuscript cites those outputs through nearby % **Source**: comments so numerical claims can be traced back to concrete artifacts.

This reproducibility layer is subordinate to the protocol claim. Numerical claims should trace to `results/`; methodological claims should trace to the definitions and protocol sections; and claim-grade statements should trace to the ledger and falsification tables. The purpose is not to make the repository a second object of study. It is to prevent a convenient output table from hiding object declarations, selection rules, robustness surfaces, or claim grades.

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